

Research statement

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My research focuses on innovation economics, examining how firms and individuals make R&D decisions and how policy shapes innovation outcomes. My work combines theoretical modeling with large-scale empirical analysis, spanning patent data, human capital allocation, and policy evaluation. Across my projects, I aim to understand the mechanisms that drive innovation and the broader economic consequences of knowledge and talent allocation.

1. Current Work Papers

1.1 Job Market Paper: Knowledge: Gift or Burden of Innovation?

My job market paper examines how knowledge accumulation affects innovation. While traditional literature highlights the benefits of accumulated knowledge in facilitating innovation through recombination, recent trends in USPTO patents show a decline in breakthrough innovations despite increasing volumes. I explore this paradox by distinguishing between field-level knowledge accumulation and direct citation of prior works, proposing that while knowledge reserves facilitate innovation, citation practices can increase the difficulty of innovating. Using a self-constructed dataset, I develop a theoretical model showing that accumulated knowledge can complement citations in reducing innovation costs, but that citation behavior may lead to diminishing returns. Empirical results support the model, indicating that citations can constrain rather than promote innovation, contributing to the observed decline in breakthrough innovations.

1.2 Hard Problems: Young Math Talents in the U.S. and Their Choices of Higher Education

My second project investigates how prestigious math competitions predict academic careers. Using a dataset of USAMO and Putnam awardees from 1985 to 2010, I track their education and career paths. I find that nearly half of the USAMO awardees obtained a Ph.D. in mathematics-related STEM fields, compared to 58.49% of Putnam awardees. Moreover, USAMO gold medalists were more likely than Honorable Mention recipients to pursue higher degrees and remain in academia. Similar patterns emerge for Putnam awardees, though relative rankings within colleges are not predictive. These findings contribute to discussions on talent screening, suggesting that competitive challenges identify strong candidates, but psychological factors and institutional environments shape long-term outcomes.

1.3 How does “Green Factory” Certification Influence Firms’ Green Innovation Performances in China? (With Jiawei Chen, Jun Ma, and Kai Wei)

My third project evaluates the impact of China's “Green Factory” policy on firm-level green innovation. Using firm-level data from 2017 onward, I assess how the policy, which promotes sustainable manufacturing, affects green patents, R&D investment, and technology transfer efficiency. Results show that “Green Factory” designation significantly increases green patents,

R&D investment, and transfer efficiency, particularly for smaller firms and non-state-owned enterprises, highlighting the heterogeneous impact of policy interventions on innovation performance.

2. Current Publications

I have published work on the impacts of the COVID-19 pandemic on China's green bond market, providing insights into the resilience of sustainable finance in times of crisis (Yi et al., 2021). Additionally, another publication on the influence of local leaders' famine experiences on housing sector development in China, exemplifies my commitment to understanding the long-term effects of historical events on economic decision-making (Hou et al., 2025).

3. Future Research Directions

3.1 Knowledge Accumulation and Innovation Networks

Extending my job market paper on knowledge accumulation and citation behavior, I aim to investigate how heterogeneity across firms and industries influences innovation outcomes. Specifically, I plan to study differences between large incumbents and startups in the accumulation and use of knowledge, as well as cross-country comparisons to evaluate how institutional environments and knowledge networks affect breakthrough innovation. Additionally, I am interested in developing dynamic models of knowledge flow using collaboration and citation networks, capturing how ideas interact across fields.

3.2 Talent Allocation and Career Trajectories

Building on my research on young mathematical talents, I plan to study how the allocation of highly skilled individuals across disciplines and institutions affects innovation outputs and long-term research productivity. I am particularly interested in integrating institutional incentives and psychological factors into empirical models of career choice, and in quantifying their influence on the production and diffusion of knowledge. By analyzing large-scale datasets of academic and professional trajectories, this work will provide insights into the determinants of talent deployment and its impact on economic growth and innovation ecosystems.

Together, these directions allow me to continue integrating theoretical modeling with large-scale empirical analysis, while maintaining a coherent research agenda across innovation, human capital, and policy.

References

Hou, Linke, Pinghan Liang, and Siyuan Lyu. "Stay hungry, and stay calm in upbeat time: Local leaders' early-life famine experience and housing sector development in China." *Journal of Regional Science* 65, no. 2 (2025): 324-338.

Yi, Xing, Caiquan Bai, Siyuan Lyu, and Lu Dai. "The impacts of the COVID-19 pandemic on China's green bond market." *Finance Research Letters* 42 (2021): 101948.